Heterophily-Aware Representation Learning on Heterogeneous Graphs

Jintang Li, Zheng Wei, Yuchang Zhu, Ruofan Wu, Huizhe Zhang, Liang Chen, and Zibin Zheng

***Abstract*—Real-world graphs are typically complex, exhibiting heterogeneity in the global structure, as well as strong heterophily within local neighborhoods. While a growing body of literature has revealed the limitations of graph neural networks (GNNs) in handling homogeneous graphs with heterophily, little work has been conducted on investigating the heterophily properties in the context of heterogeneous graphs. To bridge this research gap, we identify the heterophily in heterogeneous graphs using *metapaths* and propose two practical metrics to quantitatively describe the levels of heterophily. Our empirical investigations on real-world heterogeneous graphs have revealed that heterogeneous graph neural networks (HGNNs), which inherit many mechanisms from GNNs designed for homogeneous graphs, struggle to generalize to heterogeneous graphs with heterophily or low levels of homophily. To address the challenge, we present HETERO^{2}NET, a heterophily-aware HGNN that incorporates masked metapath prediction and masked label prediction tasks to effectively and flexibly handle both homophilic and heterophilic heterogeneous graphs. We evaluate the performance of HETERO^{2}NET on five real-world heterogeneous graph benchmarks with varying levels of heterophily. Experimental results demonstrate that HETERO^{2}NET outperforms strong baselines in the semi-supervised node classification task. In particular, HETERO^{2}NET scales to an industrial-scale commercial graph with 13 M nodes and 157 M edges, demonstrating its effectiveness in handling large and complex heterogeneous graphs.**

***Index Terms*—Heterogeneous graphs, heterogeneous graph neural networks, homophily and heterophily.**

I. INTRODUCTION

GRAPHS have become increasingly prevalent in real-world applications due to their ability to model complex relationships and interconnected structures across various domains. Building on this momentum, graph neural networks (GNNs) have emerged as a powerful framework for analyzing graph-structured data, achieving state-of-the-art performance across a wide range of graph learning tasks [1], [2], [3], [4]. In recent years, researchers have actively explored the potential of GNNs in handling heterogeneous graphs [5], [6], [7]. Heterogeneous graphs, also known as heterogeneous information networks, are network structures that comprise multiple types of nodes and edges. Each node or edge represents a different type of entity or relationship. The multitude of node and edge types in heterogeneous graphs poses great challenges in exploring rich and diverse semantics on and between nodes.

In response to the challenge of *heterogeneity*, numerous heterogeneous graph neural networks (HGNNs) have been proposed to address the relevant tasks, including heterogeneous node classification, link prediction, and recommendation [8], [9]. As an extension of GNNs in the context of heterogeneous graphs, HGNNs can jointly learn structural and semantic information from heterogeneous graphs. One line of research in HGNNs is to define and employ metapaths to model heterogeneous structures [5], [8], [10], [11]. Another line of research involves metapath-free paradigms that aggregate messages from a node's local neighborhood like GNNs but use extra modules to embed semantic information, such as node types and edge types into propagated messages [6], [7], [12], [13], [14].

In parallel with the success of GNNs and HGNNs, a growing body of literature [15], [15], [16], [17], [18], [19], [20] has revealed a significant limitation: common GNNs often exhibit poor performance on heterophilic (particularly homogeneous) graphs, where linked nodes may possess dissimilar labels and/or attributes ("opposites attract" principle). While existing HGNNs inherit many mechanisms from GNNs designed for homogeneous graphs, it remains unclear whether HGNNs can effectively handle heterogeneous graphs with heterophily, which have not yet been properly defined and explored in literature.

To bridge this research gap, this paper aims to establish a new foundation by providing a proper definition of heterophily on heterogeneous graphs via *metapaths* - a well-defined concept on heterogeneous graphs [5], [8]. As illustrated in Fig. 1, we identify three fundamental challenges in learning over heterogeneous graphs: heterogeneity, label heterophily, and attribute heterophily. In this context, heterophily refers to the presence of dissimilarities or differences between two nodes of the same type that are connected by a metapath. To provide a quantitative description of the levels of heterophily, we introduce two practical metrics: *metapath-based label homophily (MLH)* and *metapath-based Dirichlet energy (MDE)*. MLH extends the concept of homophily ratio [16] from homogeneous graphs to heterogeneous ones, while MDE is defined based on the

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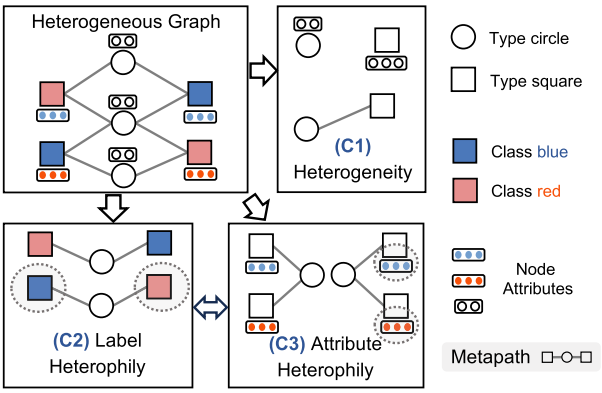


Fig. 1. The illustration of three challenges in real-world graphs: **C1** - complex structure with heterogeneous types of nodes and edges; **C2 & C3** - heterophilic labels & attributes that each node type associates with.

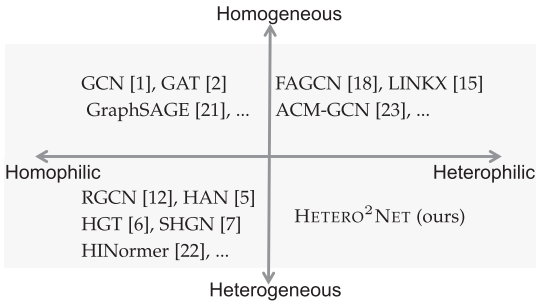


Fig. 2. Comparison of representative methods across different graph contexts. HETERO^{2}NET focuses on the challenges of heterogeneous graphs with heterophily, an area that has been less explored.

Dirichlet energy to measure the smoothness of features in local neighborhoods. Through our empirical studies, we observe that HGNNs struggle to generalize to heterogeneous graphs with heterophily (or low/medium level of homophily), as measured by MLH and MDE (Section IV-B). In such scenarios, we also find that even models that ignore the graph structure altogether, such as multilayer perceptrons (MLPs), can outperform HGNNs in many cases.

In light of the above challenges, in this work we present HETERO^{2}NET, a novel heterophily-aware HGNN to handle both homophilic and heterophilic heterogeneous graphs in terms of labels or attributes. Technically, HETERO^{2}NET adopts masked metapath prediction to jointly learn disentangled homophilic and heterophilic representations, respectively. Both representations capture different graph attribute signals that facilitate downstream tasks. In addition, HETERO^{2}NET adopts masked label prediction to enhance message propagation among nodes, especially those that exhibit strong label heterophily. Both designs contribute to addressing attribute and label heterophily, respectively.

Our contributions are summarized as follows:

- *Heterophily Measures & Current Limitations:* We introduce two metapath-based metrics, MLH and MDE, which measure the label and attribute heterophily of heterogeneous graphs, respectively. Our empirical experiments reveal the limitation of HGNNs to learn over heterogeneous graphs with heterophily, which has been largely ignored in the literature due to the lack of well-defined measures.
- *New Model for Heterophily:* We are motivated to propose HETERO^{2}NET, a novel heterophily-aware HGNN in response to the challenge of learning on heterogeneous yet heterophilic graphs (technical contributions are highlighted in Fig. 2). By incorporating both disentangled masked graph prediction and masked label prediction tasks, HETERO^{2}NET can effectively and flexibly handle both homophilic and heterophilic heterogeneous graphs.
- *Extensive Empirical Evaluations:* We conduct experiments on five real-world heterogeneous graphs, including an industrial-scale commercial graph with 13 M nodes and 157 M edges. HETERO^{2}NET demonstrates superior performance and scalability compared to strong baselines in the semi-supervised node classification task.

Paper organization: The rest of the paper is organized as follows. Section II reviews existing studies related to our work. Section III introduces the basic notations and some necessary preliminaries. Section IV introduces two metapath-based metrics and presents an empirical study on heterogeneous graphs with varying levels of heterophily. Then, we propose our HETERO^{2}NET framework in Section V, which includes disentangled masked metapath prediction and masked label prediction techniques. Experimental results are presented in Section VI. Finally, we conclude the paper in Section VII and highlight emerging research directions for the broader application of heterogeneous graph learning.

II. Related Work

In this section, we provide a review of relevant literature that is closely related to our work, including heterogeneous graph neural networks and graph representation learning over heterophilic graphs.

A. Heterogeneous Graphs Neural Networks

In the past few years, a broad range of works on HGNNs has emerged to meet the increasing demand for effectively modeling various heterogeneous graphs [23]. According to the way to deal with different graph semantics, HGNNs can be broadly categorized into metapath-based and metapath-free methods.

Metapath-based HGNNs propagate and aggregate neighbor features using hand-crafted or automatically selected metapaths. For example, Metapath2Vec [8] adopts metapath-guided random walks to capture the semantic information of heterogeneous nodes. GTN [24] assigns learnable weights to different metapaths to automatically learn useful metapaths. HAN [5] leverages hierarchical attention to describe node-level and semantic-level structures. As a follow-up work, MAGNN [11] improves HAN by introducing metapath-based aggregation to learn semantic messages from multiple metapaths.

Metapath-free HGNNs extend message passing and aggregation of GNNs to heterogeneous graphs without manually designed meaningful metapaths. RGCN [12] and its follow-up

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RGAT [25] propose to capture relation-specific patterns for each edge type with GNNs and then fuse different semantic information together. SHGN [7] incorporates a multi-layer graph attention network and extends the edge attention with a learnable edge-type embedding. HGSL [26] jointly learns the heterogeneous graph structure and the GNN parameters by exploiting complex interactions. Inspired by the success of Transformer [27], HGT [6] and HINormer [22] propose to incorporate self-attention into the graph-based message passing mechanism for modeling the structural dependencies among heterogeneous nodes.

Despite the progress in HGNNs, few methods explore the heterophily problem in heterogeneous graphs. We note that a recent work [28] seeks to address this challenging problem; however, it only scratches the surface by focusing on label heterophily and lacks scalability for large-scale heterogeneous graphs.

B. Learning Over Graphs With Heterophily

Graphs with heterophily, where connected nodes are likely to possess distinct attributes or labels, have been receiving growing attention in the research community [15], [15], [16], [17], [18], [19], [20]. The heterophily issue poses unique challenges in graph analysis and modeling. Pei et al. [16] first pay attention to this property and provides a metric to measure the homophily level of a graph. Subsequently, Zhu et al. [15] further investigate the performance degradation of GNNs when learning from graphs with heterophic or non-homophilic characteristics. They proposed H_2GCN to enhance learning from the graph structure in the presence of heterophily. CPGNN [17] generalizes GNNs to handle graphs with both homophily and heterophily by incorporating a compatibility matrix. FAGCN [18] introduces an adaptive integration mechanism for low-frequency and high-frequency signals during message passing. LINKX [15] focuses on scaling GNNs to large-scale heterophilic graphs by leveraging MLPs to learn and fuse feature matrix and adjacency matrix information. Luan et al. [20] analyze heterophily from the perspective of post-aggregation node similarity and propose a multi-channel mixing mechanism to extract rich localized information for diverse node heterophily situations.

However, the majority of research has been conducted on homogeneous graphs, leaving the heterophily problems largely unexplored in the context of heterogeneous graphs. Given the growing prevalence of heterogeneous graphs, the demand for effective methods to handle heterophily and tackle the challenges it poses becomes increasingly crucial. Recently, several concurrent works have been proposed to address heterogeneous graph learning in heterophilous settings [29], [30], [31], [32]. Nevertheless, existing approaches often lack systematic metrics for quantifying heterophily and a principled framework to disentangle attribute-based and label-based heterophily, which motivates our work.

III. Preliminary

In this section, we introduce the basic concepts and notations *w.r.t.* heterogeneous graphs, as well as metrics for measuring graph homophily/heterophily. The frequently used notations across this manuscript are summarized in Table I.

TABLE I
Frequently Used Notations in This Paper

Symbol	Description
$\mathcal{G}, \mathcal{V}, \mathcal{E}$	Heterogeneous graph with the set of nodes and edges
$\mathcal{A}, \mathcal{R}$	Node and edge types
$\psi(u, v)$	A specific relation between nodes u and v
$\phi(v)$	Type of node v
$\mathcal{Y}_{\text{train}}$	Set of partially observed class labels
$\mathcal{V}_{\text{train}}$	Set of labeled training nodes
$\mathcal{N}_u(\phi(v))$	Set of neighbors of node u that are of type $\phi(v)$
$h_u^{(l)}$	Fused representation of node u at layer l
$\hat{h}_u^{(l)}$	Aggregated representation of node u at layer l
$\hat{h}_u^{(\text{homo})}$, $\hat{h}_u^{(\text{hetero})}$	Homophilic and heterophilic representations of node u
$\mathcal{P}^*$	Pre-defined metapaths in the graph (for masking)
ρ^+, ρ^-	Sets of positive and negative metapaths
$\mathcal{G}_{\text{mask}}$, $\mathcal{G}_{\text{vis}}$	Masked and its remaining visible graph
f_θ	Metapath decoder parameterized by θ

A. Heterogeneous Graphs

A heterogeneous graph or heterogeneous information network is defined as $\mathcal{G} = \{\mathcal{V}, \mathcal{E}, \mathcal{A}, \mathcal{R}, \phi, \psi\}$, where $\mathcal{V}$ and $\mathcal{E}$ represent the set of nodes and edges, respectively. A heterogeneous graph $\mathcal{G}$ is associated with a node type mapping function $\phi : \mathcal{V} \rightarrow \mathcal{A}$ and correspondingly an edge type mapping function $\psi : \mathcal{E} \rightarrow \mathcal{R}$, where $\mathcal{A}$ denotes the set of possible object types and $\mathcal{R}$ denotes the set of relations (edge types), particularly $|\mathcal{A}| + |\mathcal{R}| > 2$. Note that $\mathcal{G}$ refers to a homogeneous graph when $|\mathcal{A}| = |\mathcal{R}| = 1$. In most cases, $\mathcal{G}$ is attributed, meaning that each node $u \in \mathcal{V}$ is associated with an d-dimensional attribute vector $x_u \in \mathbb{R}^d$. In the node classification task setting, certain node types or specific nodes are associated with a class label y. However, in heterogeneous graphs, not all node types have labels, and even within labeled node types, only a subset of nodes may have annotations.

B. Metapath in Heterogeneous Graphs

Metapath: A metapath of length n is denoted as $\mathcal{P} \triangleq A_1 \xrightarrow{R_1} A_2 \xrightarrow{R_2} \cdots \xrightarrow{R_n} A_{n+1}$ (abbreviated as $A_1 A_2 \cdots A_{n+1}$), where $A_i \in \mathcal{A}$ and $R_i \in \mathcal{R}$, describing a composite relation $R = R_1 R_2 \cdots R_n$ between node types A_1 and A_{n+1}. For example, a metapath of "author $\longrightarrow$ paper $\longrightarrow$ author" in an academic heterogeneous graph indicates the co-author relationship. Typically, a metapath contains complex semantics and high-order relations, which is considered a vital component for exploring heterogeneous graphs.

Relating heterogeneous graphs to homogeneous graphs: Given a metapath $\mathcal{P} = A_1 A_2 \cdots A_{n+1}$, we can construct the corresponding metapath induced subgraph $\mathcal{G}_\mathcal{P}$, which satisfies that edge $u \in A_1 \rightarrow v \in A_{n+1}$ exists in $\mathcal{G}_\mathcal{P}$ if and only if there is at least one length-n path between u and v following the metapath $\mathcal{P}$ in the original graph $\mathcal{G}$. One step further, let $A_1 = A_{n+1}$, we have an induced homogeneous graph $\mathcal{G}_\mathcal{P}$ built upon a metapath with the end nodes are with the same type.

C. Homophily and Heterophily

Homophily metrics: The homophily metric is a way to measure the extent of correlation between certain types of graph

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attributes and adjacency structure, with the most widely adopted definition being the edge homophily, which is defined as the average agreement over adjacent nodes' label pairs [15]:

$$\mathcal{H}_{\text{edge}}(\mathcal{G}) = \frac{1}{|\mathcal{E}|} \sum_{(u,v)\in\mathcal{E}} \mathbb{1}(y_u = y_v). \tag{1}$$

The definition (1) is defined on the *graph*-level, and might not be ideal for a fine-grained explanation of GNN performance when the underlying task is on the *node*-level, which is also of main interest in this paper. Therefore, we additionally adopt another useful definition that is essentially a reweighted version of (1) via considering the node-wise average of averaged neighborhood agreement, which is also referred to as node homophily [20], [33]:

$$\mathcal{H}_{\text{node}}(\mathcal{G}) = \frac{1}{|\mathcal{V}|} \sum_{u\in\mathcal{V}} \mathcal{H}(u) \triangleq \frac{1}{|\mathcal{V}|} \sum_{u\in\mathcal{V}} \frac{\sum_{v\in\mathcal{N}_u} \mathbb{1}(y_u = y_v)}{|\mathcal{N}_u|}. \tag{2}$$

An advantage of the definition (2) is that it enables assessing the correlation of node-wise tasks with local homophily metrics $\mathcal{H}(u)$.

Despite their popularity, a notable caveat of label-related homophily definitions is that they may be ill-defined when the underlying graph is label-scarce, which is often encountered in practice. There have been recent proposals that generalize the definition of homophily beyond label information [20], [34], which treat homophily as a measure of attribute smoothness over the underlying graph. Motivated by such generalizations, we propose to use *Dirichlet energy* regarding node features as another measure of homophily. Specifically, we begin with the definition in [35]:

$$\mathscr{E}_{\text{edge}}(\mathcal{G}) = \frac{1}{4} \sum_{(u,v)\in\mathcal{E}} \left\| \frac{\mathrm{x}_u}{\sqrt{|\mathcal{N}_u|}} - \frac{\mathrm{x}_v}{\sqrt{|\mathcal{N}_v|}} \right\|_2^2. \tag{3}$$

Analogously, we may generalize definition (3) to its node-level counterpart $\mathscr{E}_{\text{node}}$ by computing the node-wise average of local energies $\mathscr{E}(u)$, following similar philosophy as described in (2). We omit it in this context for the sake of simplicity. A smaller Dirichlet energy is regarded as indicating stronger evidence of attribute-level homophily.[1] The introduction of two notions of homophily is somewhat necessary since the performance of GNN is affected by the distributions of both attributes and labels, which are separately captured by the two metrics introduced.

Heterophily ≠ heterogeneity: In line with the focus of this work, we remark that the terms heterophily and heterogeneity are distinct concepts. Heterophily, formally defined, refers to the tendency of nodes with different attributes or labels to be connected in a graph. It emphasizes the existence of connections between nodes with contrasting characteristics. On the other hand, heterogeneity refers to the diversity or variability of nodes and edges within the graph. It is worth noting that a graph can be both homophilic and heterophilic, regardless of its heterogeneity.

[1] By saying *small*, we are implicitly assuming that the scale of $\mathscr{E}_{\text{edge}}(\mathcal{G})$ or $\mathscr{E}_{\text{node}}(\mathcal{G})$ is controlled within a proper range, which might not be the case for energy-based definitions unless the feature matrix is appropriately normalized [35]. In the subsequent explorations, we will empirically verify that the actual scale of these metrics does not affect our findings. Therefore, we postpone a more detailed discussion regarding the scale of Dirichlet energies to Section III-D.

D. On the Scale of Dirichlet Energies

In this section, we elaborate more on using Dirichlet energies as a measure of homophily. We will start from the scale of (3) on homogeneous graphs, extensions to metapath-induced graphs are straightforward. The definition is closely related to the renowned Laplacian quadratic form with respect to the normalized Laplacian [35] by

$$\mathscr{E}_{\text{edge}}(\mathcal{G}) = \frac{1}{2}\text{trace}\left(\mathbf{X}^T\mathbf{L}\mathbf{X}\right), \tag{4}$$

where $\mathbf{L} = \mathbf{I} - \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}$ is the normalized graph Laplacian. The above definition suggests that if the scale of the feature matrix X is properly controlled, then the scale of the energy will be well-controlled by the variational definition of eigenvalues:

$$\mathscr{E}_{\text{edge}}(\mathcal{G}) \le \frac{\|\mathbf{L}\|_{\text{op}}}{2} \le 1 \quad \text{if } \|\mathbf{X}\|_2 \le 1. \tag{5}$$

Where we use $\|\cdot\|_{\text{op}}$ to denote the operator norm. In practice, the feature matrix might have a Frobenius norm greater than 1, resulting in energies larger than 1. In practice, the energy of actual graphs (or graph views) is well controlled by some moderate constants, which alleviates the need to conduct normalization procedures that might make the graph learning procedure much harder due to the distortion of the feature scale.

Next, we discuss node-level notions of Dirichlet energy which is written as:

$$\begin{aligned}\mathscr{E}_{\text{node}}(\mathcal{G}) &= \frac{1}{|\mathcal{V}|}\sum_{u\in\mathcal{V}} \mathscr{E}(u) \\ &= \frac{1}{|\mathcal{V}|}\sum_{u\in\mathcal{V}} \frac{1}{4|\mathcal{N}_u|} \sum_{v\in\mathcal{N}_u} \left\| \frac{x_u}{\sqrt{|\mathcal{N}_u|}} - \frac{x_v}{\sqrt{|\mathcal{N}_v|}} \right\|_2^2. \end{aligned} \tag{6}$$

To obtain a scale upper bound on $\mathscr{E}(u)$, it suffices to use 1 as a neighborhood size lower bound and apply Cauchy-Schwartz to arrive at $\mathscr{E}(u) \le \frac{M}{2}$, where M is an ℓ_2-norm upper bound on *per-node* feature. This bound is much more realistic than controlling the Frobenius norm of the feature matrix, thereby justifying our choice of adopting node-level metrics. As we see from Section IV-B, M is well controlled in real-world datasets by some moderate constants, and once again we need no extra normalization procedures for our empirical evaluations.

IV. Empirical Investigation on Heterophily

In this section, we perform empirical investigations on how representative HGNNs behave on heterogeneous graphs with heterophilic labels and attributes to support our claims.

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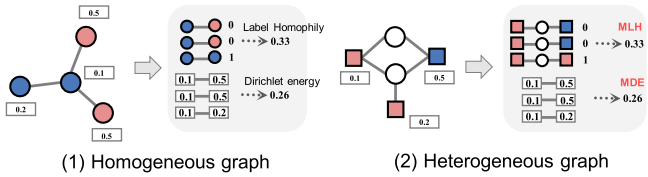


Fig. 3. Illustration of homophily metrics and our proposed heterogeneous metrics (MLH and MDE) on homogeneous and heterogeneous graphs, respectively.

A. Metapath-Based Homophily Metrics

Recall that in Section III-C, we introduced two metrics to measure the label and attribute homophily of a graph, respectively. However, these metrics were originally defined for homogeneous graphs and cannot be directly applied to heterogeneous graphs, as their computation requires both endpoints of an edge (u, v) to belong to the same label space $\mathcal{Y}$ or attribute space $\mathcal{X}$ (i.e., $y_u, y_v \in \mathcal{Y}$ or $x_u, x_v \in \mathcal{X}$), which is typically only valid in homogeneous settings. Moreover, in some cases, certain nodes belonging to specific types may lack label annotations or node attributes. In this regard, traditional homophily metrics are ill-defined for edges connecting nodes of different types, and it is not appropriate to ignore node types during computation.

This limitation has motivated us to introduce two homophily metrics that extend the definition of homophily ratio and Dirichlet energy to heterogeneous graphs via metapath. Formally, for a given node type $A \in \mathcal{A}$, we define the *metapath-based label homophily* (MLH) and *metapath-based Dirichlet energy* (MDE) as follows:

$$\begin{aligned} \text{MLH}_A(\mathcal{G}) &= \frac{1}{|\mathscr{P}|} \sum_{\mathcal{P} \in \mathscr{P}} \mathcal{H}(\mathcal{G}_{\mathcal{P}}), \\ \text{MDE}_A(\mathcal{G}) &= \frac{1}{|\mathscr{P}|} \sum_{\mathcal{P} \in \mathscr{P}} \mathscr{E}(\mathcal{G}_{\mathcal{P}}), \end{aligned} \tag{7}$$

where $\mathscr{P} = \{A \dots A_i \dots A | A_i \in \mathcal{A}\}$ represents the set of all possible metapaths starting from node type A and ending at node type A. Both $\mathcal{H}$ and $\mathscr{E}$ can be either edge-level or node-level metrics. Note that computing MLH and MDE with longer metapaths introduces significantly higher computational complexity, as the number of possible metapaths grows exponentially with their length. Since existing homophily metrics for homogeneous graphs typically rely on direct neighbors (one-hop neighbors), we consider only length-2 metapaths for the computation of both metrics. We further discuss and empirically justify this choice in Section VI-D. Typically, a larger MLH or a smaller MDE suggests a stronger homophily of a heterogeneous graph.

To better illustrate both metrics for homogeneous and heterogeneous graphs, we provide an illustration in Fig. 3. The computation of MLH and MDE shares similar insights with their counterparts on metapath-induced homogeneous graphs, which further clarifies why MLH and MDE serve as natural and effective extensions for quantifying heterophily in heterogeneous graphs.

B. Empirical Study

Experimental setup: To investigate how HGNNs behave on heterogeneous yet heterophilic graphs, we design semi-supervised node classification experiments on two heterogeneous graph datasets, DBLP and MAG, which have different scales and heterophily. Dataset statistics are listed in Table II. The experiments are performed on three representative HGNNs covering metapath-based and metapath-free architectures, including HAN [5], HGT [6], and SHGN [7]. All HGNNs follow the hyperparameter settings in their original papers. In addition, we include a multilayer perceptron (MLP) as a baseline for comparison, which does not utilize any structural information from the graph. We calculate the MLH and MDE over disjoint subgraphs that are partitioned by local node-level homophily metrics, and examine how these models perform on nodes with different levels of heterophily. We report the average performance of Micro-F1 scores over 5 runs. The Micro-F1 scores on nodes with different MLH and MDE on DBLP and MAG are illustrated in Fig. 4.

Observations: From Fig. 4, we can make the following observations: (i) Both MLH and MDE metrics demonstrate strong correlations with the performance of HGNNs, revealing a large discrepancy between nodes with varying MLH and MDE values. Specifically, HGNNs tend to achieve better performance on

TABLE II
DATASET STATISTICS

	DBLP	ACM	IMDB	MAG	RCDD
#Nodes	26,128	10,942	21,420	1,939,743	13,806,619
#Edges	239,566	547,872	86,642	21,111,007	157,814,864
#Node Types	4	4	4	4	7
#Edge Types	6	8	6	4	7
#Classes	4	3	5	349	2
Tr. nodes	30%	30%	28%	94%	0.7%
Target	author	paper	movie	paper	item
MLH (Label) ↓	0.87	0.60	0.24	0.21	0.45
MDE (Attri.) ↑	0.80	0.74	0.59	0.04	0.16

Both MLH and MDE are calculated over the target node type, and darker color indicates stronger heterophily in terms of labels or attributes. Tr. nodes refers to the ratio of training nodes.

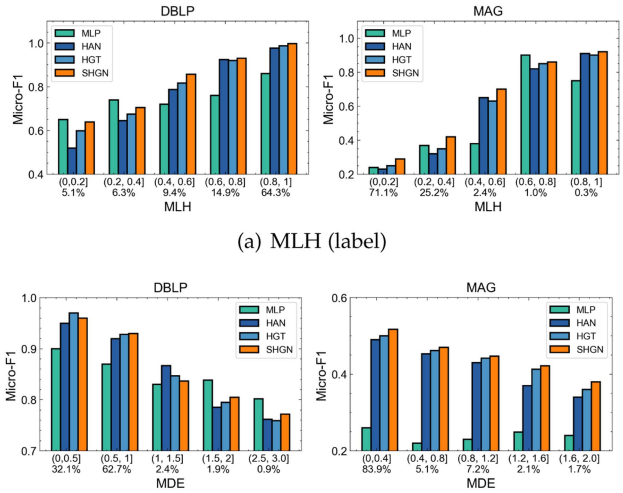


Fig. 4. Performance of MLP and different HGNNs on nodes with varying MLH and MDE values, respectively. The proportion of nodes within each bin is indicated at the bottom of the axis.

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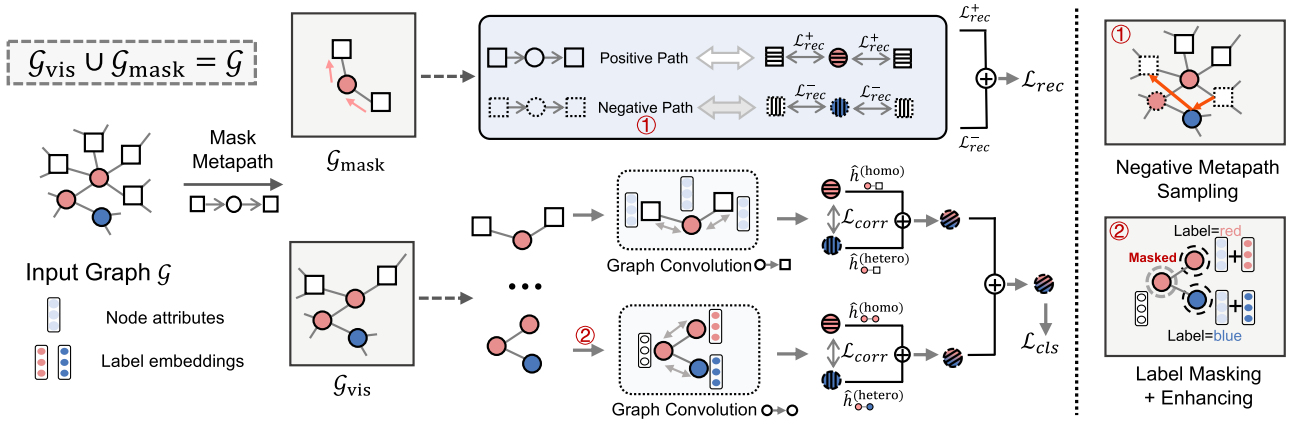


Fig. 5. The overall framework of HETERO^{2}NET. HETERO^{2}NET is a semi-supervised learning framework with two auxiliary tasks: (i) masked metapath prediction and (ii) masked label prediction. In particular, the goal of HETERO^{2}NET is to learn disentangled homophilic and heterophilic representations from individual views of the heterogeneous graph by predicting the masked metapaths and the masked node labels.

nodes that exhibit strong homophily in their local structure, as indicated by larger values of MLH or smaller values of MDE. (ii) HGNNs fail to generalize effectively to nodes with strong heterophily. In particular, a simple MLP can even outperform HGNNs on such nodes in most cases. (iii) In the DBLP dataset, which exhibits high label homophily, the performance gaps between nodes with different MLH values are relatively smaller. Conversely, in the MAG dataset, the performance gaps between nodes with different MLH values are larger. We can observe a similar trend revealed by the MDE metric as well.

In conclusion, both MLH and MDE provide valuable insights for investigating the heterophilic property in heterogeneous graphs. The heterophily issue poses a significant challenge for HGNNs, yet it has been largely overlooked in current literature. This highlights the need for further research efforts to address the heterophily issue on heterogeneous graphs effectively.

V. METHODOLOGY

In this section, we present HETERO^{2}NET as a heterophily-aware representation learning framework for heterogeneous graphs. As shown in Fig. 5, HETERO^{2}NET first extends the message passing scheme from homogeneous graphs to heterogeneous ones from a multi-view fusion perspective (Section V-A). Then, HETERO^{2}NET incorporates two key designs to address the challenges posed by label and attribute heterophily, including (i) disentangled masked metapath prediction (Section V-B) and (ii) masked label prediction (Section V-C). While several designs have been utilized separately in some prior works [36], [37], we are the first to discuss their importance under heterophily by providing extensive empirical analysis on a variety of heterogeneous graph datasets. In what follows, we will give the details of the proposed HETERO^{2}NET framework.

A. Multi-View Graph Fusion

Representation learning on heterogeneous graphs can be approached from a multi-view graph fusion perspective [12]. This involves first performing message passing within individual graph views and then conducting graph fusion across multiple views. Specifically, a heterogeneous graph composed of $|\mathcal{R}|$ types of relations can be regarded as a multiplex graph, consisting of $|\mathcal{R}|$ individual views. Each view represents a bipartite graph that focuses on a specific relationship type between nodes. For each graph view formulated by a relation $\psi(u, v)$ with u as the source node and v as the target node, we perform the well-established *message passing* scheme in which each representation of node u is computed recursively by aggregating representations from its immediate neighbors that are of a specific type $\phi(v)$. Formally, given L layers, the updating process of the l-th layer in each graph view could be formulated as:

$$\hat{h}_u^{(l)}(\psi(u,v)) = \textsc{Com}\left(\left\{\textsc{Ag}\left(\left\{h_v^{(l-1)}(\phi(v)) : v \in \mathcal{N}_u(\phi(v))\right\}\right)\right\}\right), \tag{8}$$

where $\textsc{Ag}(\cdot)$ is the *aggregation* function that aggregates representations from the neighbors (in some way—e.g., average), and $\textsc{Com}(\cdot)$ denotes the *combination* of aggregated features from a central node and its neighbors. Both AG and COM can be followed by a non-linear transformation with learnable parameters. $\mathcal{N}_u(\phi(v))$ is the set of u's neighbors that is of type $\phi(v)$. $h_u^{(l)}(\cdot)$ is the (aggregated) representation of node u at the l-th layer and initially $h_u^{(0)} = x_u$.

Then, the representations from different views that correspond to the same node type are fused or merged to obtain the node representation:

$$h_u^{(l)}(\phi(u)) = \textsc{Fuse}\left(\left\{\hat{h}_u^{(l)}(\psi(u,v)) | \psi(u,v) \in \mathcal{R}\right\}\right), \tag{9}$$

where $\textsc{Fuse}(\cdot)$ denotes the *fusion* on representations of the same semantics learned from different views. Typically, $\textsc{Fuse}(\cdot)$ is implemented as one of the following functions: MEAN, SUM, or CONCAT. In particular, RGCN [12] is the special case with SUM the fusing function. Note that each graph view in the heterogeneous graph has its own set of graph convolution layers with learnable parameters. This means that the overall model, with a depth of L, consists of $L \times |\mathcal{R}|$ layers.

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B. Disentangled Masked Metapath Prediction

Early work on learning with heterophilic graphs has demonstrated that low-frequency information, specifically homophilic signals, plays a crucial role in the success of GNNs [18], [38]. Moreover, heterophilic signals, which capture the difference between nodes, are also vital for effectively handling heterophilic graphs and mitigating the oversmoothing problem. Although both homophilic and heterophilic signals are important, learning and determining appropriate signals to use from a graph can be challenging due to the complex correlation between downstream tasks and different types of information.

In this work, we resort to disentangled representation learning, which has been widely used to learn underlying representations that disentangle hidden factors in the graph structure. Our goal is to decouple the *homophilic* and *heterophilic* representations from each graph view and explicitly represent the graph with the two disentangled representations:

$$\hat{h}_u \triangleq \hat{h}_u^{(\text{homo})} + \hat{h}_u^{(\text{hetero})}, \tag{10}$$

where $\hat{h}_u^{(\text{homo})}$ and $\hat{h}_u^{(\text{hetero})}$ are homophilic and heterophilic representations of node u, respectively. Here $\hat{h}_u^{(\text{homo})}$ and $\hat{h}_u^{(\text{hetero})}$ are learned from two different channels of the same graph convolution layer, as stated in (8). For the sake of notation simplicity, we omit the relation $\psi(u,v)$ from $\hat{h}_u^{(\cdot)}(\psi(u,v))$ hereafter. Particularly, $\hat{h}^{(\text{homo})}$ and $\hat{h}^{(\text{hetero})}$ should exhibit two desired *properties*:

- *P1: Disentanglement*: $\hat{h}^{(\text{homo})}$ and $\hat{h}^{(\text{hetero})}$ are disentangled from each other. This means that they are explicitly separated and represent distinct aspects of the graph's information.
- *P2: Semantic-preserving*: $\hat{h}^{(\text{homo})}$ captures homophilic signals in the node's local neighborhood, while $\hat{h}^{(\text{hetero})}$ captures heterophilic signals. In other words, $\hat{h}^{(\text{homo})}$ focuses on encoding the similarities and community structure within a node's neighborhood, while $\hat{h}^{(\text{hetero})}$ emphasizes the differences and dissimilarities between nodes.

To encourage node disentanglement *(P1)* between latent representations $\hat{h}^{(\text{homo})}$ and $\hat{h}^{(\text{hetero})}$, we have to enforce the statistical independence between them. It is noteworthy that if two variables X and Y are statistically independent, then we have $\mathbb{E}(X,Y) = \mathbb{E}(X)\mathbb{E}(Y)$. Therefore, the minimization of the correlation between $\hat{h}^{(\text{homo})}$ and $\hat{h}^{(\text{hetero})}$ could be used to achieve the disentanglement between homophilic and heterophilic representations. In this work, we adopt a Pearson's correlation coefficient as the correlation loss between $\hat{h}^{(\text{homo})}$ and $\hat{h}^{(\text{hetero})}$, i.e.,

$$\mathcal{L}_{\text{corr}} = \sum_{u\in\mathcal{V}} \frac{\left|\text{Cov}\left(\hat{h}_u^{(\text{homo})}, \hat{h}_u^{(\text{hetero})}\right)\right|}{\sqrt{\text{Var}\left(\hat{h}_u^{(\text{homo})}\right)}\sqrt{\text{Var}\left(\hat{h}_u^{(\text{hetero})}\right)}}, \tag{11}$$

where $\text{Cov}(\cdot,\cdot)$ and $\text{Var}(\cdot)$ indicate *covariance* and *variance* operations, respectively. The Pearson correlation coefficient measures linear dependence between two representations. In (11), the correlation coefficient between $\hat{h}_u^{(\text{homo})}$ and $\hat{h}_u^{(\text{hetero})}$ is encouraged to converge to 0. Actually, as the correlation loss converges, the homophilic representations $\hat{h}_u^{(\text{homo})}$ and the heterophilic representations $\hat{h}_u^{(\text{hetero})}$ are statistically independent [39].

Although the goal of disentangling $\hat{h}^{(\text{homo})}$ and $\hat{h}^{(\text{hetero})}$ can be achieved by minimizing the correlation loss $\mathcal{L}_{\text{corr}}$, it is necessary to introduce an additional task to assign semantic meaning to these representations *(P2)*. In other words, we aim to associate homophily and heterophily with $\hat{h}^{(\text{homo})}$ and $\hat{h}^{(\text{hetero})}$, respectively. In this work, we extend the homophily assumption from homogeneous graphs[2] to heterogeneous graphs from the perspective of metapath, i.e.,

> *Nodes connected by a metapath are more likely to share homophilic properties, such as classes or attributes:*

Based on the above assumption, it is natural to learn $\hat{h}^{(\text{homo})}$ by enforcing the proximity between nodes and their local neighborhoods. There are several alternative tasks available in the literature, but in this work, our focus is on masked graph autoencoding. Masked graph autoencoding has emerged as a promising self-supervised task and has demonstrated remarkable performance in various graph learning tasks [36], [37].

Inspired by the recent work MaskGAE [36], which focuses on masked edge/path reconstruction in homogeneous graphs, our work shifts its focus to masked graph modeling in heterogeneous graphs. In this subsection, we introduce a novel and principled task that utilizes masked metapath prediction to learn meaningful representations. Given a set of pre-defined metapaths $\mathcal{P}^* = \{\mathcal{P}_1, \mathcal{P}_2, \ldots\}$, the input heterogeneous graph is masked by a set of adjacent paths $\rho^+ = \{u_1u_2\cdots u_{\text{n}} | \phi(u_1)\phi(u_2)\cdots\phi(u_{\text{n}}) \in \mathcal{P}^* \wedge (u_i,u_j)\in\mathcal{E}\}$. Let $\mathcal{G}_{\text{mask}}$ denote the graph masked from $\mathcal{G}$ (with ρ^+ masked) and $\mathcal{G}_{\text{vis}}$ the remaining visible graph. our goal is to learn $h^{(\text{homo})}$ by reconstructing $\mathcal{G}_{\text{mask}}$ using the remaining visible structure $\mathcal{G}_{\text{vis}}$. Given that two nodes disconnected would potentially share heterophilic information, we can also learn $h^{(\text{hetero})}$ by reconstructing the negative metapaths $\rho^- = \{u_1u_2\cdots u_{\text{n}} | \phi(u_1)\phi(u_2)\cdots\phi(u_{\text{n}}) \in \mathcal{P}^* \wedge (u_i,u_j)\notin\mathcal{E}\}$. Here the negative metapaths are defined by a sequence of nodes that match the type pattern of specific metapath in $\mathcal{P}^*$ but are disconnected in the heterogeneous graph. Overall, we optimize the following binary cross-entropy loss:

$$\begin{aligned}
\mathcal{L}_{\text{rec}}^{+} &= \frac{1}{|\rho^+|}\sum_{(u,v)\in\rho^+} \log f_\theta(h_u^{(\text{homo})}, h_v^{(\text{homo})}),\\
\mathcal{L}_{\text{rec}}^{-} &= \frac{1}{|\rho^-|}\sum_{(u,v)\in\rho^-} \log f_\theta(h_u^{(\text{hetero})}, h_v^{(\text{hetero})}),\\
\mathcal{L}_{\text{rec}} &= -\left(\mathcal{L}_{\text{rec}}^{+} + \mathcal{L}_{\text{rec}}^{-}\right),
\end{aligned} \tag{12}$$

where f_θ is a standard two-layer MLP parameterized by θ, which is defined as:

$$f_\theta(h_u,h_v) = \mathbf{W}_1\cdot\text{ReLU}(\mathbf{W}_0\cdot(h_u\circ h_v)), \quad \theta=\{\mathbf{W}_0,\mathbf{W}_1\}, \tag{13}$$

where $\circ$ denotes the element-wise product, $\mathbf{W}_0$ and $\mathbf{W}_1$ represent learnable matrices of MLP. We omit the bias terms in the

[2] In literature, it is fundamentally assumed that nodes with similar attributes, characteristics, or properties are more likely to be connected in the graph.

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MLP for simplicity, although they can be easily incorporated if desired. By predicting the missing metapaths within the graph, our model is encouraged to capture and understand the underlying relationships and patterns between nodes of different types.

Comparison with prior work: Previous disentangled techniques typically adopt a multichannel graph convolutional layer to promote node disentanglement among diverse latent representations [9], [39], [40], [41]. These methods intend to enforce independent capture of isolated hidden factors by each dimension of the representation in each channel. However, channel-wise or dimension-wise disentanglement might not be efficient in training, especially as the number of channels and dimensions grows. Consequently, disentangled representation learning struggles to scale effectively to large and heterogeneous graphs. In contrast, we present a simple yet effective disentangled way utilizing only two independent channels, i.e., homophilic and heterophilic channels. One step further, we avoid the need for dimension-wise disentanglement through decorrelation and the masked metapath prediction task. This enables HETERO^{2}NET to overcome the scalability issue when dealing with large real-world heterogeneous graphs.

C. Masked Label Prediction

While disentangled representation learning with masked metapath prediction is effective in handling heterogeneous graphs with heterophily, particularly in node attributes, it does not directly leverage label information during message propagation. This limitation may potentially lead to unsatisfied performance, especially in graphs with strong label heterophily.

Inspired by [42], we resort to the label propagation algorithm [43] to solve the label heterophily issue. Label propagation (LP) is one of the classic message passing algorithms in semi-supervised node classification, which predicts a class for each node by propagating the known labels in the graph. The classic LP typically assumes that two connected nodes are more likely to have the same class, and thus iteratively propagate the class labels from labeled nodes to their unlabeled neighbors. To adapt LP to graphs with heterophily directly, several methods have been explored in the literature, such as employing pseudo labels [44], compatibility matrix [45], or an additional learnable weight matrix [46]. Although LP can be effective in leveraging the graph structure to infer labels for unlabeled nodes, it is typically used as a standalone component rather than being explicitly integrated into the message passing design.

In this work, we propose to incorporate the partially observed labels into message passing by propagating both ground truth labels and node features through the given graph. Given partially observed class labels, denoted as $\mathcal{Y}_{\text{train}} = \{y_0, y_1, \ldots\}$, where each $y_i \in \mathbb{R}^{C\times 1}$ represents a one-hot vector encoding the class label of a labeled node. We first map the one-hot label vectors $\mathcal{Y}_{\text{train}}$ into a continuous space using a linear projection. Next, we add the mapped labels into the node features as follows:

$$\hat{x}_u = x_u + \mathbf{W}_y \cdot y_u, \tag{14}$$

where $\mathbf{W}_y$ is a learnable weight matrix that embeds $\mathcal{Y}_{\text{train}}$ into the same space as node features. By combining the label information with the existing features, we create augmented node representations that capture both the original features and the label information.

To prevent the leakage of label information during training, we adopt a masked label prediction strategy inspired by masked graph autoencoding [36]. During each training round, we randomly mask a portion of node labels following a specific distribution, e.g., Bernoulli distribution:

$$\begin{aligned} m &\sim \text{Bernoulli}(p), \\ \hat{x}_u &= x_u + \mathbf{W}_y \cdot (y_u \circ m), \end{aligned} \tag{15}$$

where $0 < p \leq 1$ is the masking ratio for the node labels. In this way, the augmented node representations are propagated through the graph, allowing the label information to be shared and refined across neighboring nodes without leakage. During the inference stage, we set the masking ratio p to 0, which ensures that the model can leverage the complete label information $\mathcal{Y}_{\text{train}}$ to guide its predictions.

D. Overall Objective Function

As a general graph learning framework, HETERO^{2}NET follows the standard semi-supervised node classification pipeline and employs cross-entropy loss for optimization:

$$\mathcal{L}_{\text{cls}} = \sum_{u\in\mathcal{V}_{\text{train}}} \text{CROSSENT}(\tilde{y}_u, y_u), \tag{16}$$

where CROSSENT is the cross-entropy loss and $\mathcal{V}_{\text{train}}$ is the set of labeled training nodes. Note that, for multi-label node classification, we can employ binary cross-entropy as the optimization loss.

Integrating the correlation loss in (11), the reconstruction loss in (12), with the semi-supervised node classification loss (16), the overall objective function of HETERO^{2}NET is formulated as:

$$\mathcal{J} = \mathcal{L}_{\text{cls}} + \alpha\mathcal{L}_{\text{corr}} + \beta\mathcal{L}_{\text{rec}}, \tag{17}$$

where $\alpha \geq 0$ and $\beta \geq 0$ are non-negative hyperparameters trading off three terms. To help better understand the proposed framework, we provide the detailed algorithm for training HETERO^{2}NET in Algorithm 1.

E. Time Complexity of HETERO^{2}NET

In this subsection, we discuss the time complexity of HETERO^{2}NET in the full-batch and mini-batch learning settings, respectively. Essentially, HETERO^{2}NET is a set of GNNs defined over $|\mathcal{R}|$ graph views, and each view undergoes the step of *message passing*. Assuming we have L layers, the *full-batch* training and inference time complexity for each graph view with L layers can be bounded by $\mathcal{O}(L|\mathcal{E}|d + L|\mathcal{V}|d^2)$. Here, $\mathcal{O}(L|\mathcal{E}|d)$ represents the total cost of sparse-dense matrix multiplication or message passing, and $\mathcal{O}(L|\mathcal{V}|d^2)$ represents the total cost of linear transformation on node representations. We assume a hidden dimension of d across all layers for simplicity. For large-scale graphs, we can perform neighborhood sampling, which involves sampling a subset of the neighbors at each layer to reduce the neighborhood size. Let s be the number

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Algorithm 1: Heterophily-Aware HGNN (HETERO^{2}NET).

Input: The input graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}, \mathcal{A}, \mathcal{R}, \phi, \psi\}$, node attributes $\{x_u, \forall u \in \mathcal{V}\}$, pre-defined mathpath set $\mathcal{P}^* = \{\mathcal{P}_1, \mathcal{P}_2, \ldots\}$, hyperparameters α, β, and the number of layers L;
Output: The prediction probability $\{\hat{y}_u, \forall u \in \mathcal{V}\}$;

while *not converged* **do** ;
  // Mask the input graph based on metapaths $\mathcal{P}^*$
  $\mathcal{G}_{\text{mask}}$ and $\mathcal{G}_{\text{vis}} \leftarrow \mathcal{G}$;
  for $u \in \mathcal{V}$ **do**
    // Mask and generate enhanced features;
    $\hat{x}_u = x_u + \mathbf{W}_y \cdot (y_u \circ m), m \sim \text{Bernoulli}(p)$;
    // Multi-view graph fusion
    $h_u^0 = \hat{x}_u$;
    for $\psi(u, v) \in \mathcal{R}$ **do**
      Calculate $h_u^{(l)}(\psi(u, v))$ by (8);
      Calculate $h_u^{(l)}(\phi(u))$ by (9);
    end for
    $\hat{h}_u = \hat{h}_u^{(\text{homo})} + \hat{h}_u^{(\text{hetero})}$;
    Calculate correlation loss $\mathcal{L}_{\text{corr}}$ by (11);
    Calculate reconstruction loss $\mathcal{L}_{\text{rec}}$ by (12);
    Calculate classification loss $\mathcal{L}_{\text{cls}}$ by (16);
    Calculate $\mathcal{J} = \mathcal{L}_{\text{cls}} + \alpha\mathcal{L}_{\text{corr}} + \beta\mathcal{L}_{\text{rec}}$;
  end for
end while ;
$\hat{y}_u \leftarrow \text{SOFTMAX}(h_u^{(L)}), \forall u \in \mathcal{V}$;

of sampled neighbors for each node, and b be the batch size. The total per-epoch cost of HETERO^{2}NET is then bounded by $\mathcal{O}(|\mathcal{V}|s^L d + |\mathcal{V}|s^{L-1}d^2)$. In most cases, s and L are typically small, making the overall time complexity of HETERO^{2}NET acceptable for large-scale scenarios.

VI. EXPERIMENTS

In this section, we perform experimental evaluations to demonstrate the effectiveness of our proposed HETERO^{2}NET framework.

A. Experimental Setups

Datasets: In our experiments, we consider a total of five real-world heterogeneous graphs: three academic citation datasets, including DBLP [7], ACM [5], and MAG [47]; one movie-related dataset, IMDB [7]; and one commodity-related dataset, RCDD [48]. The detailed statistics of all datasets are shown in Table II. We present the heterogeneous graph structure in Fig. 6. For all datasets, we use public train/valid/test splits for the node classification task. The overall homophily metrics for each dataset can be calculated by averaging the corresponding metapath-wise metrics, as shown in (7). Based on the empirical analysis of the MLH and MDE metrics, we select all length-2 metapaths in which both the starting and ending node types match the target classification node type as the reconstruction

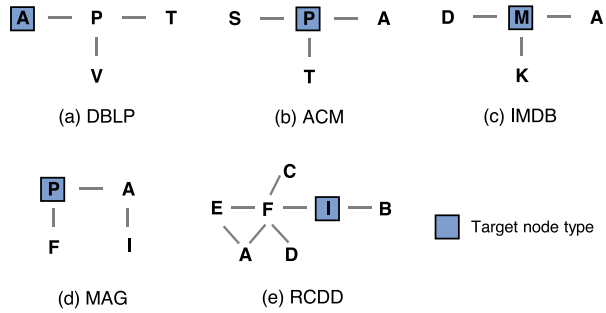


Fig. 6. Illustration of the heterogeneous graph structure for each dataset.

objective. The ablation study on longer metapaths is presented in Section VI-D. Here are the details of the datasets:

- *DBLP*[3] is a bibliography website focusing on computer science. It contains two node types, including author (A) and paper (P). In our experiments, we focus on the classification of authors, and the corresponding *length-2 metapath* is *APA*.
- *IMDB*[4] is a website providing movie-related information. It includes four types of nodes: movie (M), director (D), actor (A), and keyword (K). In our experiments, we focus on the classification of movies, and the corresponding *length-2 metapaths* are {*MDM*, *MAM*, *MKM*}.
- *ACM*[5] is a citation network that consists of four node types: paper (P), author (A), subject (S), and term (T). In our experiments, we focus on the classification of papers, and the corresponding *length-2 metapaths* are {*PAP*, *PSP*, *PTP*}.
- *MAG*[6] is another academic network with multiple node types, including paper (P), field of study (F), and author (A). In our experiments, we concentrate on the classification of papers, and the corresponding *length-2 metapaths* are {*PFP*, *PAP*}.
- *RCDD*[7] is a risk commodity detection dataset. Its node types consist of item (I), objects named 'f' (F), objects named 'b' (B), and others. In our experiments, we focus on the classification of items, and the corresponding *length-2 metapaths* are {*IFI*, *IBI*}.

Additional challenges in RCDD: scalability, class-imbalance, and noise: RCDD is a risk commodity detection dataset based on a real risk detection scenario from Alibaba's e-commerce platform. It is a large-scale heterogeneous graph with multiple node and edge types. It is worth noting that RCDD is a class-imbalanced dataset, where the distribution of black ($y = 0$) and white ($y = 1$) samples is severely imbalanced (approximately 10:1). In addition, RCDD is a challenging dataset with structure noise. Malicious users may forge "innocent" relationships by forging devices, addresses, or other methods. Fig. 7

[3] http://web.cs.ucla.edu/yzsun/data/
[4] https://www.kaggle.com/karrrimba/
[5] http://dl.acm.org/
[6] https://ogb.stanford.edu/docs/nodeprop/
[7] https://zenodo.org/record/8103003

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TABLE III
QUANTITATIVE RESULTS (%) ON THE SEMI-SUPERVISED NODE CLASSIFICATION TASK. THE RESULTS ARE AVERAGED OVER FIVE RUNS. THE BEST AND RUNNER-UP RESULTS ARE HIGHLIGHTED USING *BOLD* AND UNDERLINE, RESPECTIVELY. H_1: HETEROGENEITY-ORIENTED NETWORK, H_2: HETEROPHILY-AWARE NETWORK

		Datasets→	DBLP		IMDB		ACM		MAG	
H_1	H_2	Methods↓	Macro-F1	Micro-F1	Macro-F1	Micro-F1	Macro-F1	Micro-F1	Macro-F1	Micro-F1
		MLP	$80.36_{\pm 0.31}$	$80.92_{\pm 0.25}$	$57.01_{\pm 0.67}$	$61.65_{\pm 0.93}$	$86.50_{\pm 0.32}$	$86.53_{\pm 0.31}$	$6.35_{\pm 0.38}$	$23.60_{\pm 0.17}$
		GCN	$90.84_{\pm 0.32}$	$91.47_{\pm 0.34}$	$57.88_{\pm 1.18}$	$64.82_{\pm 0.64}$	$92.17_{\pm 0.24}$	$92.12_{\pm 0.23}$	$25.14_{\pm 0.33}$	$47.26_{\pm 0.36}$
		GAT	$91.05_{\pm 0.76}$	$91.73_{\pm 0.50}$	$58.94_{\pm 1.35}$	$64.86_{\pm 0.43}$	$92.26_{\pm 0.94}$	$92.19_{\pm 0.93}$	$22.94_{\pm 0.49}$	$43.79_{\pm 0.24}$
	✓	LINKX	$75.05_{\pm 1.45}$	$77.78_{\pm 1.59}$	$58.98_{\pm 0.45}$	$62.03_{\pm 0.41}$	$89.91_{\pm 1.14}$	$89.69_{\pm 1.09}$	$14.63_{\pm 0.56}$	$33.43_{\pm 0.22}$
	✓	FAGCN	$82.40_{\pm 0.28}$	$83.08_{\pm 0.28}$	$63.68_{\pm 0.56}$	$67.49_{\pm 0.25}$	$89.27_{\pm 0.62}$	$89.34_{\pm 0.78}$	$16.87_{\pm 0.41}$	$37.99_{\pm 0.95}$
	✓	ACM-GCN	$81.80_{\pm 0.32}$	$82.91_{\pm 0.14}$	**65.44**$_{\pm 0.49}$	$68.71_{\pm 0.40}$	$88.97_{\pm 0.85}$	$89.21_{\pm 0.53}$	$13.42_{\pm 0.79}$	$34.47_{\pm 0.70}$
✓		RGCN	$91.52_{\pm 0.50}$	$92.07_{\pm 0.50}$	$61.26_{\pm 0.33}$	$65.21_{\pm 0.73}$	$91.95_{\pm 0.44}$	$91.75_{\pm 0.35}$	$27.01_{\pm 0.21}$	$48.80_{\pm 0.24}$
✓		HAN	$91.67_{\pm 0.49}$	$92.05_{\pm 0.62}$	$57.74_{\pm 0.96}$	$64.63_{\pm 0.58}$	$90.89_{\pm 0.43}$	$90.79_{\pm 0.43}$	$8.94_{\pm 0.16}$	$26.76_{\pm 0.36}$
✓		RGAT	$92.61_{\pm 0.48}$	$93.15_{\pm 0.49}$	$57.85_{\pm 0.58}$	$62.79_{\pm 0.70}$	$90.03_{\pm 0.56}$	$90.40_{\pm 0.54}$	$24.76_{\pm 0.47}$	$45.29_{\pm 0.40}$
✓		HGT	$93.01_{\pm 0.23}$	$93.49_{\pm 0.25}$	$63.00_{\pm 1.19}$	$67.20_{\pm 0.57}$	$91.12_{\pm 0.76}$	$91.15_{\pm 0.71}$	$27.87_{\pm 0.30}$	$49.19_{\pm 0.63}$
✓		SHGN	$94.01_{\pm 0.24}$	$94.20_{\pm 0.31}$	$63.53_{\pm 1.26}$	$67.36_{\pm 0.57}$	$93.42_{\pm 0.44}$	$93.35_{\pm 0.45}$	$22.61_{\pm 0.40}$	$43.68_{\pm 0.71}$
✓		HINormer	$93.90_{\pm 0.17}$	$94.32_{\pm 0.15}$	$64.30_{\pm 0.92}$	$67.62_{\pm 0.52}$	$92.23_{\pm 0.27}$	$92.15_{\pm 0.28}$	$25.80_{\pm 0.41}$	$47.57_{\pm 0.61}$
✓	✓	HETERO^{2}NET	**94.03**$_{\pm 0.35}$	**94.46**$_{\pm 0.37}$	$65.37_{\pm 0.48}$	**69.61**$_{\pm 0.72}$	**94.01**$_{\pm 0.54}$	**93.91**$_{\pm 0.61}$	**33.28**$_{\pm 0.32}$	**55.13**$_{\pm 0.24}$

† The best and runner-up results are marked as **bold** and underlined, respectively.

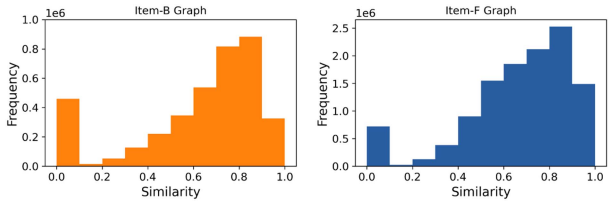


Fig. 7. Frequency of connected nodes with varying levels of similarity in RCDD. A majority of nodes with low similarity (e.g., zero) are observed to be connected in the graph, indicating the presence of structural noise.

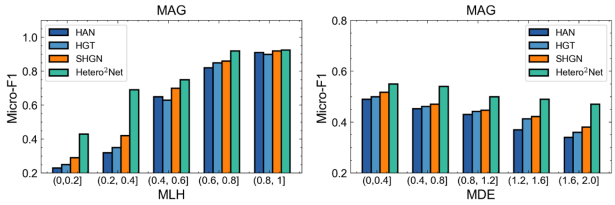


Fig. 8. Performance comparison in terms of nodes with different MLH and MDE, respectively.

illustrates the frequency distribution of pairwise similarity among connected nodes in the graph. The similarity is measured using the cosine similarity of the attributes of two connected nodes. As observed in Fig. 7, a significant portion of nodes share zero similarity, indicating the presence of noisy connections in the graph. Consequently, the RCDD dataset exhibits a high level of attribute heterophily.

Baselines: To comprehensively evaluate the proposed HETERO^{2}NET against the state-of-the-art approaches, we consider six HGNN baselines, including RGCN [12], RGAT [25], HAN [5], HGT [6], SHGN [7], and HINormer [22]. In addition, we include five homogeneous GNNs including GCN [1], GAT [2], as well as LINKX [15], FAGCN [18], and ACM-GCN [20], which are specifically designed for handling heterophilic graphs. It is important to note that we did not include a comparison with [28] due to the unavailability of open-source code for their method. The hyperparameters of all the baselines were configured according to the experimental settings officially reported by the authors and were then carefully tuned in our experiments to achieve their best results.

Evaluation settings: We evaluate the node classification performance of HETERO^{2}NET and baselines using Macro-F1 and Micro-F1 metrics. Additionally, for the RCDD dataset, we include average precision (AP) as an evaluation metric, as suggested in [48]. For reproduction, we report the averaged results with standard deviations across 5 runs. All experiments are conducted on an NVIDIA V100 GPU with 32 GB memory. Code is made publicly available at https://github.com/EdisonLeeeee/Hetero2Net.

B. Hyperparameter Settings

We set the embedding dimension of HETERO^{2}NET to 128 for small-scale datasets (DBLP, IMDB, ACM) and 256 for large-scale datasets (MAG and RCDD). The hyperparameters α and β are tuned from a set of values {0, 0.1, 0.2, 0.3, 0.4, 0.5}. The masking ratio for node labels p is tuned using a set of values ranging from 0 to 1.0. The ablation results on several key hyperparameters are shown in Section VI-E.

C. Node Classification Performance

We evaluate the effectiveness of the proposed HETERO^{2}NET through the node classification task. The results of HETERO^{2}NET and the baselines on four common benchmarks, including DBLP, IMDB, ACM, and MAG, are reported in Table III. To showcase the superiority of HETERO^{2}NET in handling industrial-scale heterogeneous graphs, we specifically highlight the experimental results on RCDD in Table IV. Additionally, Fig. 8 presents performance comparisons between HETERO^{2}NET and several representative HGNNs across nodes with varying levels of MLH

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TABLE IV
QUANTITATIVE RESULTS (%) ON THE RCDD DATASET

H_1	H_2	Methods	Macro-F1	Micro-F1	AP
		MLP	86.08$_{\pm 0.34}$	97.27$_{\pm 0.43}$	82.45$_{\pm 0.47}$
		GCN	91.30$_{\pm 0.38}$	98.29$_{\pm 0.12}$	90.85$_{\pm 0.30}$
		GAT	89.75$_{\pm 0.21}$	98.02$_{\pm 0.04}$	86.95$_{\pm 0.49}$
	✓	LINKX	OOM	OOM	OOM
	✓	FAGCN	90.62$_{\pm 0.32}$	98.09$_{\pm 0.07}$	88.95$_{\pm 0.49}$
	✓	ACM-GCN	84.89$_{\pm 0.47}$	94.12$_{\pm 0.11}$	83.21$_{\pm 0.97}$
✓		RGCN	92.25$_{\pm 0.34}$	98.30$_{\pm 0.07}$	90.73$_{\pm 0.36}$
✓		HAN	87.32$_{\pm 0.34}$	97.46$_{\pm 0.07}$	83.00$_{\pm 0.36}$
✓		RGAT	89.19$_{\pm 1.20}$	97.93$_{\pm 0.16}$	87.58$_{\pm 0.45}$
✓		HGT	91.04$_{\pm 0.48}$	98.29$_{\pm 0.08}$	88.65$_{\pm 1.03}$
✓		SHGN	88.12$_{\pm 0.52}$	97.68$_{\pm 0.10}$	83.89$_{\pm 0.73}$
✓		HINormer	OOM	OOM	OOM
✓	✓	HETERO^{2}NET	**92.92**$_{\pm 0.17}$	**98.79**$_{\pm 0.21}$	**92.87**$_{\pm 0.33}$

"OOM": out of memory on an NVIDIA V100 GPU with 32GB memory.

TABLE V
NODE CLASSIFICATION RESULTS (%) ON THE RCDD AND MAG DATASETS UNDER THE INDUCTIVE SETTING

	MAG		RCDD		
	Macro-F1	Micro-F1	Macro-F1	Micro-F1	AP
RGCN	25.59$_{\pm 0.47}$	46.38$_{\pm 0.52}$	80.80$_{\pm 0.88}$	87.85$_{\pm 0.68}$	80.28$_{\pm 0.70}$
HAN	5.58$_{\pm 0.68}$	23.40$_{\pm 0.36}$	78.76$_{\pm 0.87}$	86.90$_{\pm 0.87}$	74.44$_{\pm 0.87}$
RGAT	21.39$_{\pm 0.22}$	41.92$_{\pm 0.17}$	80.02$_{\pm 0.51}$	88.76$_{\pm 0.58}$	78.41$_{\pm 0.49}$
HGT	24.26$_{\pm 0.36}$	45.58$_{\pm 0.44}$	81.57$_{\pm 0.69}$	88.82$_{\pm 0.51}$	81.18$_{\pm 0.45}$
SHGN	18.56$_{\pm 0.30}$	39.63$_{\pm 0.27}$	78.30$_{\pm 0.78}$	87.86$_{\pm 0.82}$	74.07$_{\pm 0.74}$
HINormer	21.94$_{\pm 0.31}$	43.71$_{\pm 0.33}$	OOM	OOM	OOM
HETERO^{2}NET	**30.70**$_{\pm 0.52}$	**52.95**$_{\pm 0.38}$	**84.12**$_{\pm 0.41}$	**89.79**$_{\pm 0.32}$	**84.07**$_{\pm 0.53}$

and MDE. Based on these comprehensive evaluations, we can derive the following observations.

Overall performance: Table III demonstrates the state-of-the-art performance of HETERO^{2}NET on the node classification task. On one hand, for datasets exhibiting strong attribute heterophily, such as DBLP and ACM, HETERO^{2}NET consistently outperforms strong baselines across most cases. This indicates the effectiveness of the disentanglement module in effectively leveraging node attributes to address heterophily challenges. On the other hand, for datasets characterized by strong label heterophily, such as IMDB and MAG, HETERO^{2}NET demonstrates significant improvement over baselines. This underscores the benefit of incorporating the masked label prediction technique to enhance message propagation by leveraging label information. Also, this finding aligns with the argument presented in [42], which suggests that incorporating both node features and labels during training and inference stages can enhance model performance. It is also observed that several state-of-the-art baselines (e.g., SHGN or HINormer) achieve competitive results on small-scale datasets like DBLP, IMDB, and ACM. However, they fall significantly behind HETERO^{2}NET on large-scale datasets with complex metapath structures, particularly the MAG dataset. This is because metapaths capture meaningful semantic relationships, which become increasingly important in larger and more complex heterogeneous graphs.

Large-scale performance: To further validate the effectiveness of HETERO^{2}NET on more intricate scenarios, we conduct experiments on RCDD, an industrial-scale commercial graph comprising 13 M nodes and 157 M edges. The RCDD dataset exhibits unique challenges, including class imbalance, data noises, and multiple node and edge types, which pose difficulties for existing HGNNs to achieve satisfactory performance. We compare the performance between HETERO^{2}NET and the baselines on the RCDD dataset in Table IV. As evident from the results, most baseline methods struggle significantly with the challenges posed by scalability and data quality. In contrast, HETERO^{2}NET consistently outperforms all baselines by large margins, demonstrating its effectiveness and robustness in handling the challenges of scalability and data quality. Furthermore, an intriguing observation is that simple GCN [1] outperforms other heterophily-specific models in the majority of cases, particularly on datasets characterized by high label-wise heterophily but low attribute-wise heterophily, such as MAG and RCDD. This finding aligns with previous studies on homogeneous graphs [49], where GCN demonstrates strong performance on label-wise heterophilic graphs when the labels have distinguishable distributions.

Connections between metrics and performance: We conduct a comparison between HETERO^{2}NET and three representative HGNNs, namely HAN, HGT, and SHGN, on nodes with varying levels of heterophily within the MAG dataset. According to the results shown in Fig. 8, it is observed that HETERO^{2}NET consistently achieves superior performance on nodes that exhibit strong homophily. This observation further confirms the strong correlations between our proposed metrics and the performance of HGNNs. Remarkably, HETERO^{2}NET outperforms the baselines, especially in the case of strong attribute and label heterophily. This provides strong evidence that HETERO^{2}NET can effectively address the heterophily issue compared to existing HGNNs. Furthermore, the performance of all methods differs on nodes characterized by MLH and MDE, with larger performance gaps observed for MLH. This observation is reasonable since the MAG dataset exhibits strong label heterophily while having a relatively small attribute heterophily. Therefore, MLH and MDE can provide comprehensive and complementary insights into the impact of homophily on model performance.

Performance on inductive settings: To further demonstrate the effectiveness of HETERO^{2}NET, we conduct experiments on the MAG and RCDD datasets under the inductive setting. Specifically, we follow the standard inductive setting [21], where the model is trained on a subset of the graph and evaluated on a disjoint set of unseen nodes. We construct the training and test sets based on the original train/test splits of the graph. The inductive results on both datasets are presented in Table V, which includes only the results for HETERO^{2}NET and HGNN baselines. As shown in Table V, since we strictly follow the inductive setting, test nodes are completely unseen during training and thus are removed from the training graph. As a result, all methods experience a significant performance drop compared to the transductive scenario. Specifically, methods trained on the MAG dataset exhibit relatively smaller performance drops due to the larger number of training nodes. In contrast, the RCDD dataset, which is more challenging and contains fewer training nodes, results in a more substantial performance reduction for all methods. Among all methods, HETERO^{2}NET experiences the

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TABLE VI
ABLATION ON CORE COMPONENTS IN HETERO^{2}NET

	DBLP		IMDB		ACM		MAG	
	Macro-F1	Micro-F1	Macro-F1	Micro-F1	Macro-F1	Micro-F1	Macro-F1	Micro-F1
HETERO^{2}NET	**94.03**$_{\pm 0.35}$	**94.46**$_{\pm 0.37}$	**65.37**$_{\pm 0.48}$	**69.61**$_{\pm 0.72}$	**94.01**$_{\pm 0.54}$	**93.91**$_{\pm 0.61}$	**33.28**$_{\pm 0.32}$	**55.13**$_{\pm 0.24}$
$-\mathcal{L}_{\text{corr}}$	93.33$_{\pm 0.42}$	93.94$_{\pm 0.55}$	64.17$_{\pm 0.25}$	68.19$_{\pm 0.22}$	93.45$_{\pm 0.50}$	93.38$_{\pm 0.46}$	32.59$_{\pm 0.10}$	53.10$_{\pm 0.18}$
$-\mathcal{L}_{\text{rec}}$	92.65$_{\pm 0.65}$	93.09$_{\pm 0.47}$	64.22$_{\pm 0.32}$	68.33$_{\pm 0.59}$	93.75$_{\pm 0.44}$	93.62$_{\pm 0.53}$	31.32$_{\pm 0.16}$	52.11$_{\pm 0.15}$
$-(\mathcal{L}_{\text{corr}}\&\mathcal{L}_{\text{rec}})$	92.94$_{\pm 0.58}$	93.26$_{\pm 0.62}$	63.88$_{\pm 0.40}$	67.80$_{\pm 0.74}$	93.49$_{\pm 0.39}$	93.43$_{\pm 0.71}$	31.19$_{\pm 0.14}$	51.97$_{\pm 0.19}$
$-$MaskLP†	92.84$_{\pm 0.57}$	93.13$_{\pm 0.47}$	63.79$_{\pm 0.32}$	67.25$_{\pm 0.41}$	92.91$_{\pm 0.50}$	93.02$_{\pm 0.14}$	28.23$_{\pm 0.21}$	49.04$_{\pm 0.09}$

†MaskLP: masked label prediction.

smallest performance drop in the inductive setting. Moreover, HETERO^{2}NET consistently outperforms all HGNN baseline models on both datasets under the inductive scenario. These results demonstrate the strong inductive capability of HETERO^{2}NET, which further highlights its potential applicability in real-world industrial scenarios.

D. Ablation Study

In this subsection, we conduct extensive ablation studies on four benchmark datasets to explore the effects and alternatives of different components in HETERO^{2}NET.

We begin by evaluating the contribution of each component in HETERO^{2}NET through comparisons with several degenerate variants: (1) $-\mathcal{L}_{\text{corr}}$: we remove the correlation loss and only use the construction loss in disentanglement module; (2) $-\mathcal{L}_{\text{rec}}$: we remove the construction loss and only consider correlation loss in disentanglement module; (3) $-(\mathcal{L}_{\text{corr}}\&\mathcal{L}_{\text{rec}})$: we only maintain the classification loss and remove other losses; (4) $-$MaskLP: we remove the masked label prediction module and preserve disentangled representation learning with masked metapath prediction. Based on the results of the ablation studies presented in Table VI, several key observations can be made. Overall, both the learning tasks for disentanglement and masked label prediction consistently enhance the performance of HETERO^{2}NET across all datasets. Specifically, the integrated utilization of the two learning tasks demonstrates superior performance in most cases within the disentanglement module. Furthermore, the exclusion of the masked metapath prediction technique leads to a notably larger performance degradation for the MAG dataset compared to the other datasets, thus corroborating the observation discussed in Section VI-C.

We also perform an ablation study comparing MaskLP with traditional label propagation (LP) to address label heterophily issues. As shown in Fig. 9, MaskLP consistently outperforms traditional LP across all datasets. This result demonstrates that the mask-then-predict paradigm in MaskLP is more effective than traditional label propagation for handling label heterophily, particularly in scenarios with limited labeled data or strong heterophilous relationships. The effectiveness of MaskLP further validates our design choices and highlights its potential advantages for practical heterogeneous graph learning tasks.

Finally, we investigate the effect of metapath length on computing the MLH and MDE metrics as well as on the performance of metapath reconstruction. Recall that we select all length-2 metapaths in which both starting and ending node types match the target classification node type. The experimental results are shown in Table VII. We observe that for datasets exhibiting higher homophily (larger MLH or smaller MDE values), increasing the metapath length tends to introduce more heterophilous relationships, leading to a decrease in MLH or an increase in MDE. Conversely, for datasets with higher heterophily (smaller MLH or larger MDE), longer metapaths yield the opposite effect. Moreover, increasing metapath length does not enhance performance on smaller datasets such as DBLP and ACM, and introduces substantial computational overhead. Even though longer metapaths yield minor performance gains on large-scale datasets like MAG, the significant increase in computational complexity outweighs these benefits. Given that previous homophily metrics for homogeneous graphs rely on direct neighbors (one-hop neighbors), longer metapaths might not be appropriate for accurately measuring homophily or heterophily in heterogeneous graphs. Thus, our results suggest that

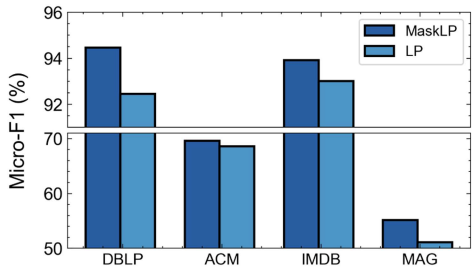


Fig. 9. Ablation on our proposed MaskLP and traditional LP.

TABLE VII
COMPARISON OF DIFFERENT METAPATH LENGTHS

		DBLP	IMDB	ACM	MAG
Length=2	MLH	0.87	0.60	0.24	0.21
	MDE	0.80	0.74	0.59	0.04
	Macro-F1	**94.03**$_{\pm 0.35}$	**65.37**$_{\pm 0.48}$	**94.01**$_{\pm 0.54}$	33.28$_{\pm 0.32}$
	Micro-F1	**94.46**$_{\pm 0.37}$	**69.61**$_{\pm 0.72}$	**93.91**$_{\pm 0.61}$	55.13$_{\pm 0.24}$
Length=4	MLH	0.80	0.52	0.42	0.29
	MDE	0.61	0.71	0.40	0.07
	Macro-F1	92.98$_{\pm 0.28}$	63.65$_{\pm 0.27}$	92.89$_{\pm 0.67}$	**33.51**$_{\pm 0.34}$
	Micro-F1	93.27$_{\pm 0.31}$	67.71$_{\pm 0.43}$	92.47$_{\pm 0.72}$	**55.24**$_{\pm 0.28}$
Length=6	MLH	0.70	0.46	0.49	0.39
	MDE	0.54	0.65	0.24	0.10
	Macro-F1	92.72$_{\pm 0.30}$	63.01$_{\pm 0.26}$	91.83$_{\pm 0.26}$	30.72$_{\pm 0.21}$
	Micro-F1	93.11$_{\pm 0.34}$	67.24$_{\pm 0.51}$	92.05$_{\pm 0.41}$	51.39$_{\pm 0.17}$

Note that we consider only metapaths of lengths 2, 4, and 6, since we require the starting and ending node types to match the target classification node type.

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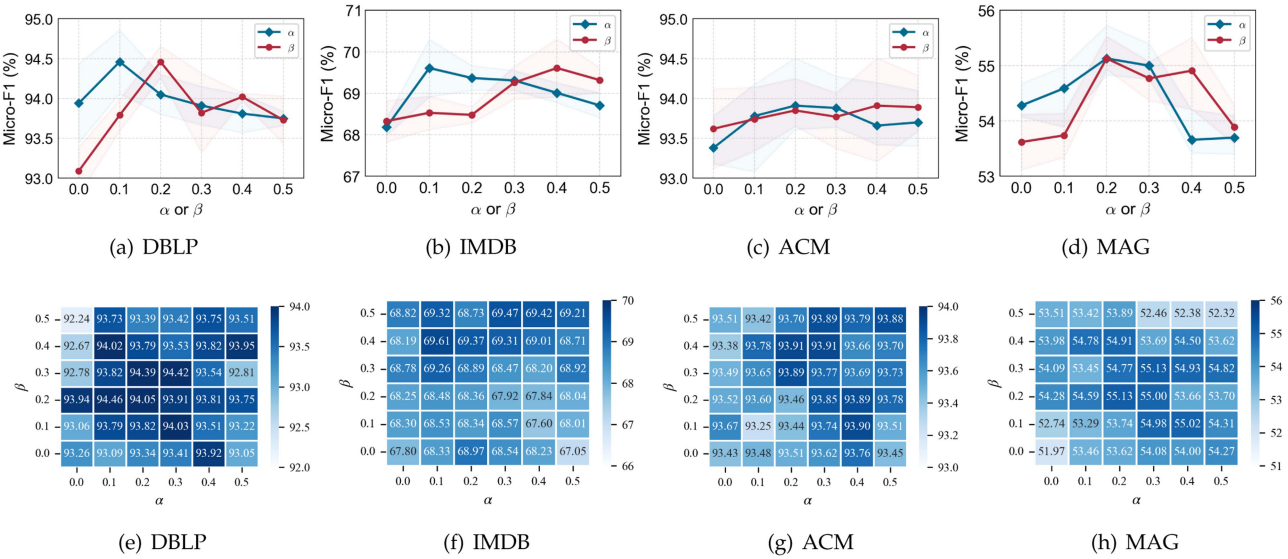


Fig. 10. Ablation on hyperparameters α and β. Experiments were conducted on DBLP, IMDB, ACM, and MAG over five runs.

length-2 metapaths provide an optimal trade-off between capturing meaningful relationships and maintaining computational efficiency.

E. Hyperparameter Sensitivity

Trade-offs α and β: To provide an in-depth analysis of them, we provide the ablation results of α and β, ranging from $\{0, 0.1, 0.2, 0.3, 0.4, 0.5\}$. The results on DBLP, IMDB, and ACM are presented in Fig. 10. In Fig. 10(a)–(d), when varying α, we fix β at its optimal value, and vice versa when varying β. This approach isolates the effect of each parameter while maintaining the model's overall performance. To further investigate the joint impact of varying both parameters simultaneously, we also present additional results in Fig. 10(e)–(h), which illustrate how simultaneous adjustments of both parameters affect the model's performance. We observe that (i) both α and β play important roles in HETERO^{2}NET. Better performance is achieved when reasonable values are assigned to α and β. (ii) HETERO^{2}NET is more sensitive to changes in β across all datasets, as increasing its value generally improves the model's performance. (iii) Larger values of α tend to result in poorer performance, as it overly emphasizes the node disentanglement and may introduce more noise to the learned representations. Overall, finding an appropriate balance between α and β is crucial for achieving optimal performance in HETERO^{2}NET.

Visualization of disentangled representations: To further verify the effectiveness of disentangled representation learning, i.e., the homophilic representations are statistically independent to heterophilic representations, we follow previous works [39] and compute the correlation between the two representations. In this context, lower correlation values indicate that the homophilic and heterophilic representations capture more distinct and non-overlapping information. To evaluate this, we implement HETERO^{2}NET on the MAG dataset and visualize the correlation matrix of the learned representations in Fig. 11(a), where the final representation is obtained by concatenating the homophilic and heterophilic components associated with "Paper" nodes (each with 8 dimensions). The resulting correlation map consists of four blocks, with the upper-right and lower-left blocks representing the cross-correlation between homophilic and heterophilic representations. As shown in Fig. 11(a), the correlations in these regions are low, indicating that the homophilic information learned by HETERO^{2}NET is well-separated from the heterophilic information.

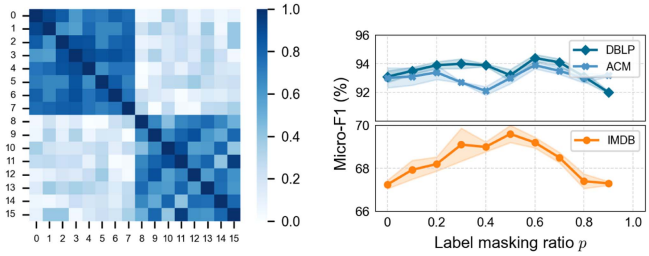


Fig. 11. (a) The correlation map of homophilic (0–7) and heterophilic (8–15) representations learned by the proposed HETERO^{2}NET on the MAG dataset; (b) Ablation on label masking ratio p.

Label masking ratio p: As masked label prediction is a core component of HETERO^{2}NET, we conduct ablation studies on the label masking ratio p for the training nodes. The ablation experiments are performed on the DBLP, IMDB, and ACM datasets. We vary p in the range from 0 to 1, with an increment of 0.1. As shown in Fig. 11(b), the results reveal a clear trend in the impact of varying p on the model's performance. Higher masking ratios generally lead to better generalization on the test set, suggesting that more challenging training conditions may encourage the model to develop more robust features. However, a ratio of 1.0, where all labels are masked, results in a slight performance decrease across all datasets. Furthermore, p is

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closely related to the structural noise or the heterophily level of the dataset and can be tuned based on the MLH metric. For datasets like DBLP and ACM, where the MLH value is relatively high (indicating lower heterophily), a smaller p is sufficient to achieve good results. Since the label distribution in local neighborhoods is more homophilous, fewer masked labels are needed to guide the learning process effectively. In contrast, for datasets like IMDB, which exhibit higher heterophily (lower MLH values), a larger p is required to achieve better performance. This is because label information is more scattered across different node neighborhoods, requiring a stronger masking effect to help the model learn meaningful label embeddings. In summary, these findings validate the significance of the masked label prediction task and emphasize the importance of selecting an appropriate value for p.

VII. Conclusion and Future Work

In this paper, we explore the heterophily property in heterogeneous graphs and introduce two metapath-based metrics, MLH and MDE, to quantify the heterophily of labels and attributes in heterogeneous graphs, respectively. Our empirical results indicate that both metrics effectively correlate with the performance of HGNNs. Furthermore, we present HETERO^{2}NET, a principled framework designed to learn from both homophilic and heterophilic heterogeneous graphs. In our comprehensive experimental evaluation, HETERO^{2}NET consistently outperforms strong baselines in the semi-supervised node classification task. Particularly, HETERO^{2}NET advances new state-of-the-art performance on an industrial graph dataset RCDD with challenging scenarios.

Learning on heterogeneous and heterophilous graphs is an important yet challenging step towards practical machine learning in real-world applications. In future work, we aim to extend HETERO^{2}NET to diverse heterogeneous graph learning tasks for enhanced adaptability and performance across various domains.

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